

2016.0 RANGE ROVER (LG), 204-06

RIDE AND HANDLING OPTIMIZATION

ACTIVE STABILIZATION SYSTEM BLEEDING (G1551263)

GENERAL PROCEDURES

60.60.13	ACTIVE CORNERING ENHANCEMENT (ACE) - SYSTEM BLEED	ALL DERIVATIVES	1.3	USED WITHINS
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SPECIAL TOOL(S)



204-591-
01

Dynamic Response
Bleed Bottle



JLR-204- 591-03

Adaptor, Dynamic
Response Bleed
Bottle

C H E C K

CAUTION:

Dynamic response system (DRS) components are manufactured to very precise tolerances. It is therefore essential that absolute cleanliness is observed when working with these components. Always install blanking plugs to any open orifices or lines. Failure to follow this instruction may result in foreign matter ingress to the DRS.

NOTES:

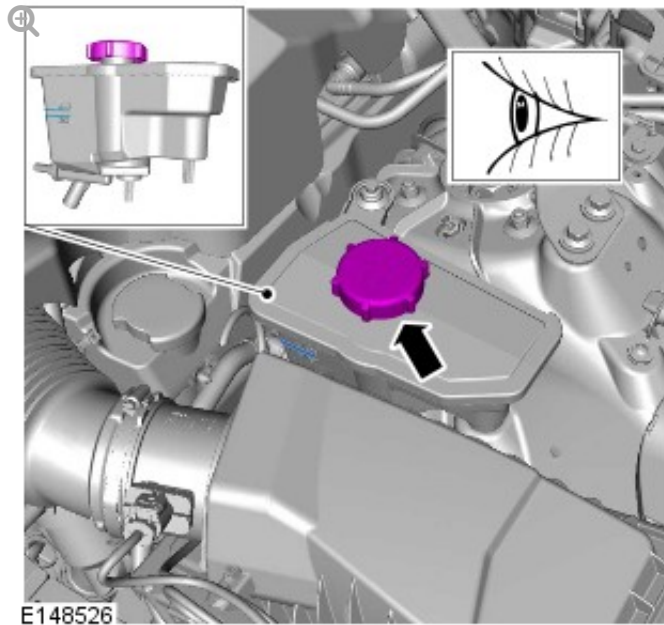
- The first part of this procedure should be carried out if the following components have been removed: suction hose, pump, high pressure hose, cooler, cooler return to reservoir. The full procedure is required if the following components have been removed: front or rear valve block to actuator pipe assemblies, or the valve block. It is possible to bleed only the front or rear of the system if only a stabilizer bar has been removed.

If the vehicle has had low fluid level and there is a possibility that the car has been driven in this condition, then air is likely to have moved into the secondary circuit and a full system bleed is required.

Within the special application for DRS in SDD here are 2 routines for bleed.

- 1. Manual bleed is the primary function that should be used in conjunction with the bleed bottle. Its function is to open the safety control valve (SCV) and proportional direction controls valves (PDCV) so that fluid can flow through the valve block to the actuators.
- 2. Auto bleed can be used to displace air that is trapped within the valves/valve block that has not been displaced by the manual bleed process, it will not remove air from the actuator or actuator pipes.
- For front actuator removal or replacement use Steps 1-11, 14-19, 24-25 and 27-31.
- For rear actuator removal or replacement use Steps 1-9, 14-15 and 20-31.
- Primary circuit bleed:

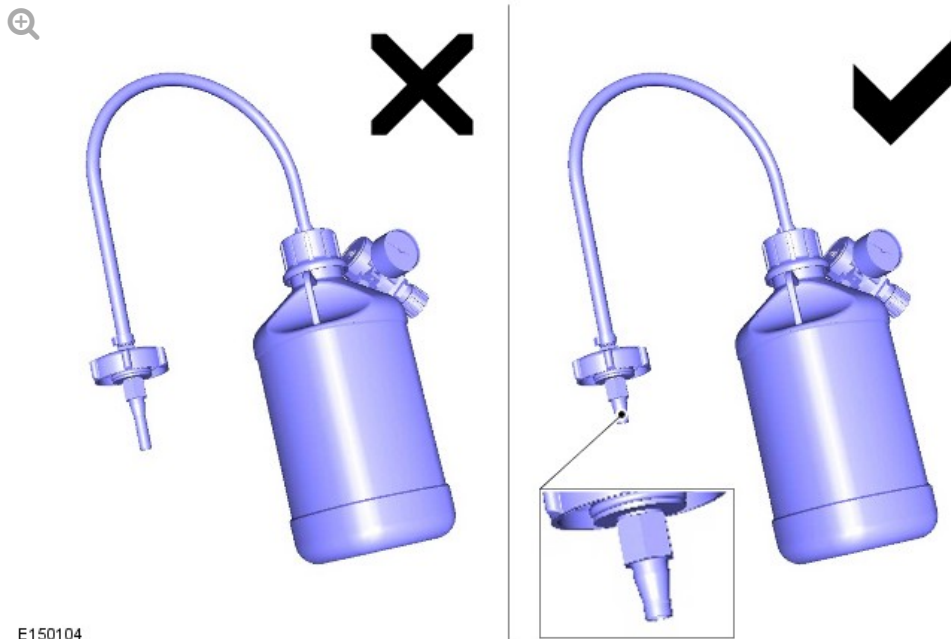
1.



Check and top-up the dynamic response system fluid reservoir.
Then squeeze the suction hose (feed hose to pump from reservoir)
several times to displace air that may be trapped under the fine
mesh of the reservoir suction side filter.

ADJUSTMENT

2.

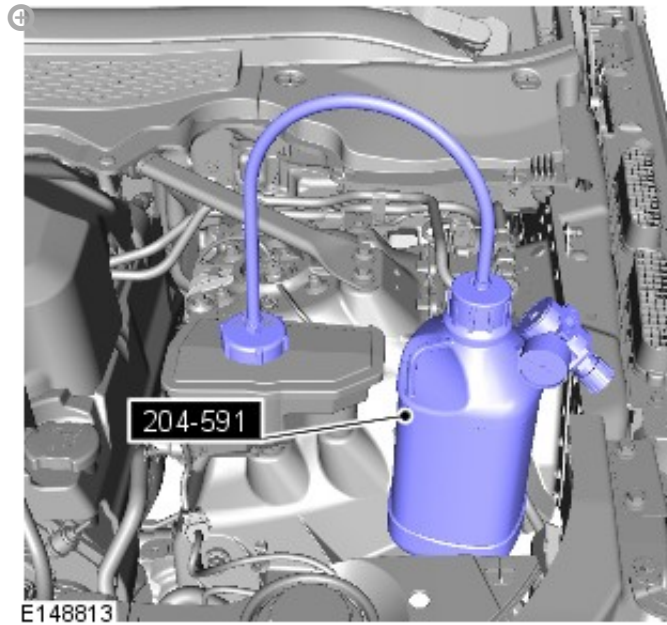


Install the new adaptor to the Dynamic response bleed bottle.

3.

NOTE:

Install the special tool to the dynamic response reservoir.
Section 2 to 4 is only required if the reservoir has been drained and or pump /suction hose has been changed.



1. Completely fill the reservoir with fluid.
2. Make sure the pressure regulator on the special tool is turned OFF.
3. Fill the special tool bottle approximately three-quarters full with fluid.
4. Connect the special tool to a suitable workshop air supply.
5. Using the special tool pressure regulator, carefully increase the air pressure to approx. 0.3 - 0.7 Bar (5 - 10 PSI).

Special Tool(s): [204-591-01](#), [JLR-204-591-03](#)

4. Start engine and run for 1 minute.

5.

WARNING:

The special tool is still pressurised when the source air pressure is removed. Release air pressure within special tool slowly before removing. Remove the special tool from the dynamic response system reservoir.

6. Check that there is fluid circulating in the reservoir and stop engine. Then squeeze the suction hose (feed hose to pump from reservoir) several times to displace air that may be trapped under the fine mesh of the reservoir suction side filter.
7. Run the engine, stopping and starting several times. Each time allow the air bubbles to settle out of the fluid in the reservoir and then squeeze the suction hose to release any trapped air. This is to encourage any air trapped in the primary circuit to be expelled by the reservoir. For suction hose, pump, high pressure hose, cooler, cooler return to reservoir, the removal and installation process is now complete. Make sure that the fluid level is set to maximum line and move to step 33 Hydraulic self-test, otherwise move to step 7.

8.

WARNING:

Make sure to support the vehicle with axle stands.

Raise and support the vehicle.

9.

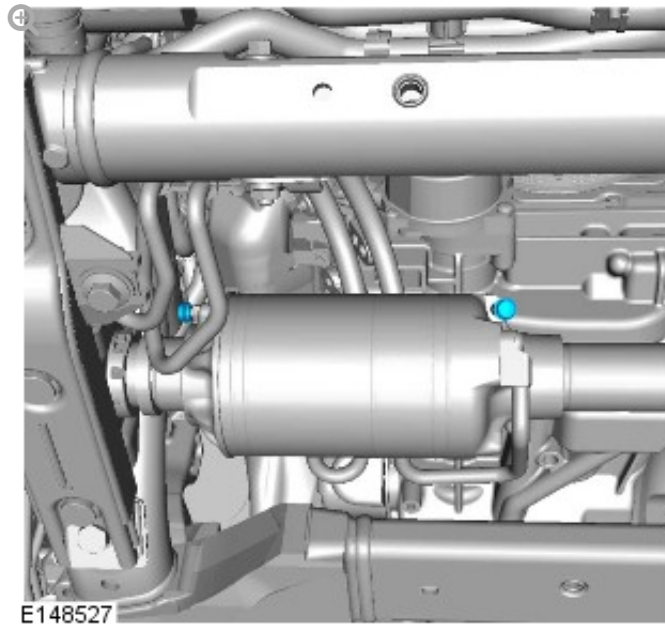
NOTE:

Follow manual bleed procedure where the following components have been removed: Valve block, front or rear valve block to actuator pipe assemblies, or actuator.

For secondary circuit bleed steps 1 to 7 must have been completed first, install / re-attach the special tool as per step 2 of adjustment.

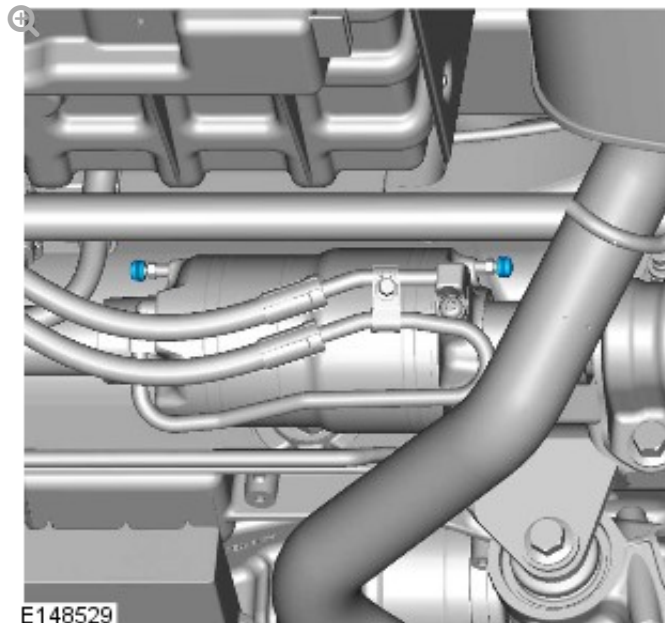
10. Attached approved diagnostic tool and navigate to dynamic response special application manual bleed.
11. Refer to: [Engine Undershield](#) (501-02 Front End Body Panels, Removal and Installation).

12.



Remove the front bleed screw covers.

13.



Remove the rear bleed screw covers.

14. For valve block or pipe removal and replacement, open all four actuator bleed screws by half a turn.
15. Start manual bleed in approved diagnostic tool, this will open the safety control valve (SCV) and move the proportional direction control valves (PDCV) to the correct position.

16.

CAUTION:

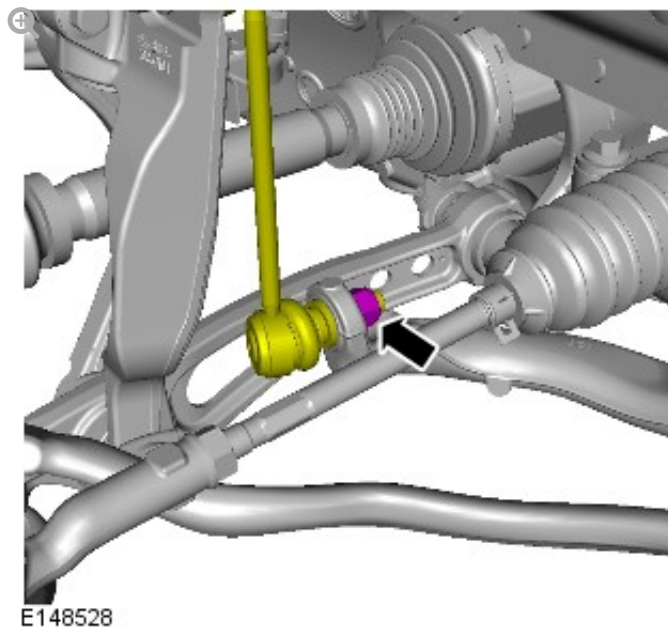
The dynamic response bleed tool fluid reservoir must remain full with new, clean fluid, at all times during bleeding.

Bleed the dynamic response system until a flow of clean, air-free fluid, is being pumped into all four bleed jars and close bleed screws.

17.

NOTE:

RH side only

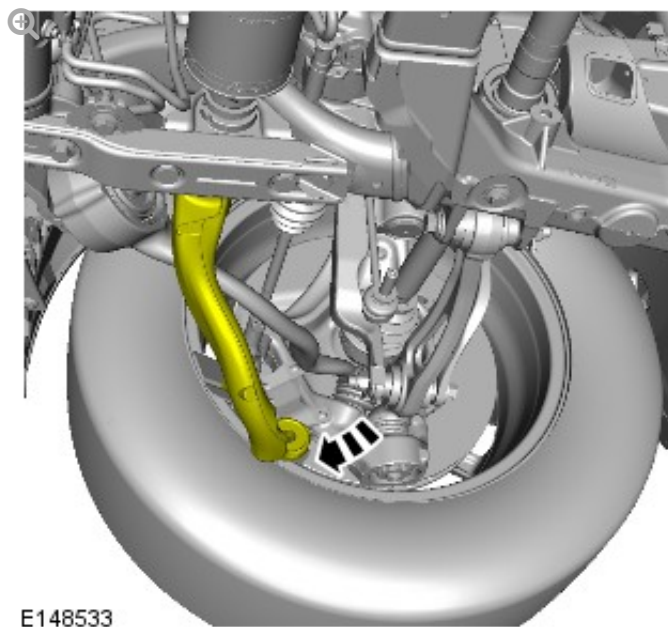


Bleed front actuator:

18.

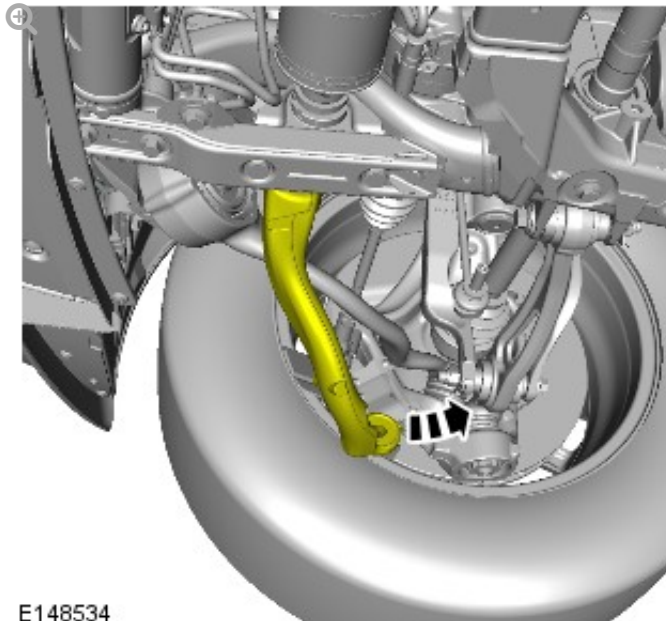
NOTE:

RH side only



Open right bleed screw and stroke right arm fully down. Allow fluid to run for a further 5 seconds. Close bleed screw.

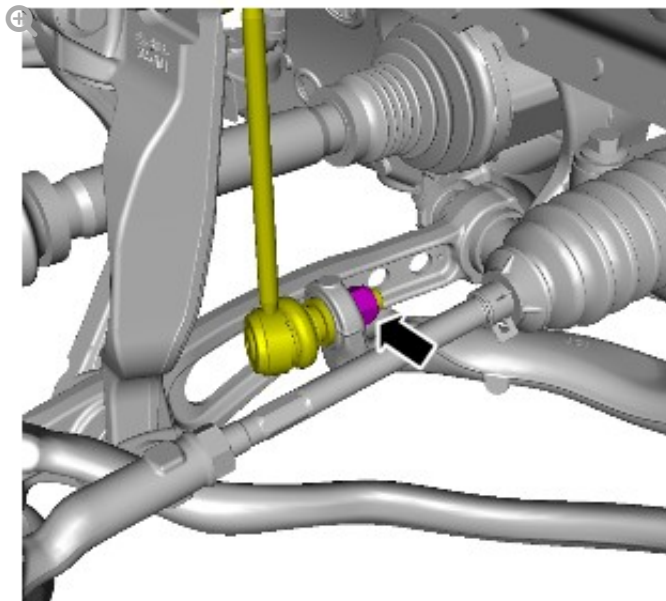
19.



E148534

Open left bleed screw and stroke right arm fully up. Allow fluid to run for a further 5 seconds. Close bleed screw.

20.



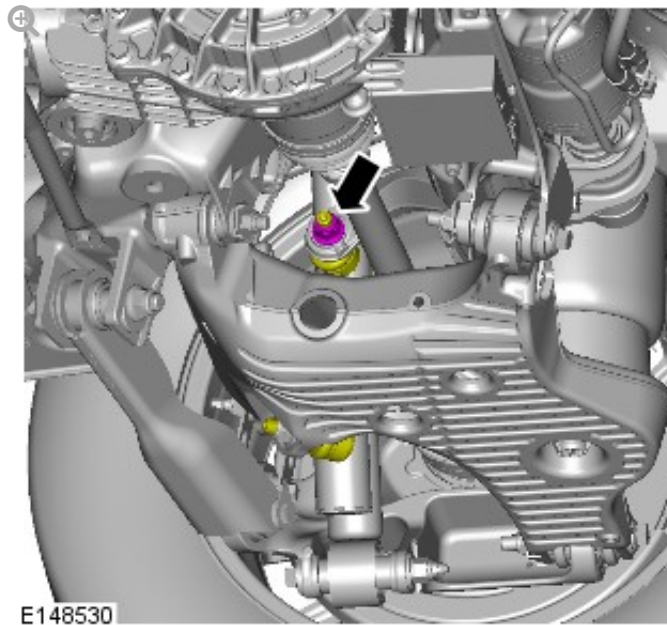
E148528

Torque: 185 Nm

21.

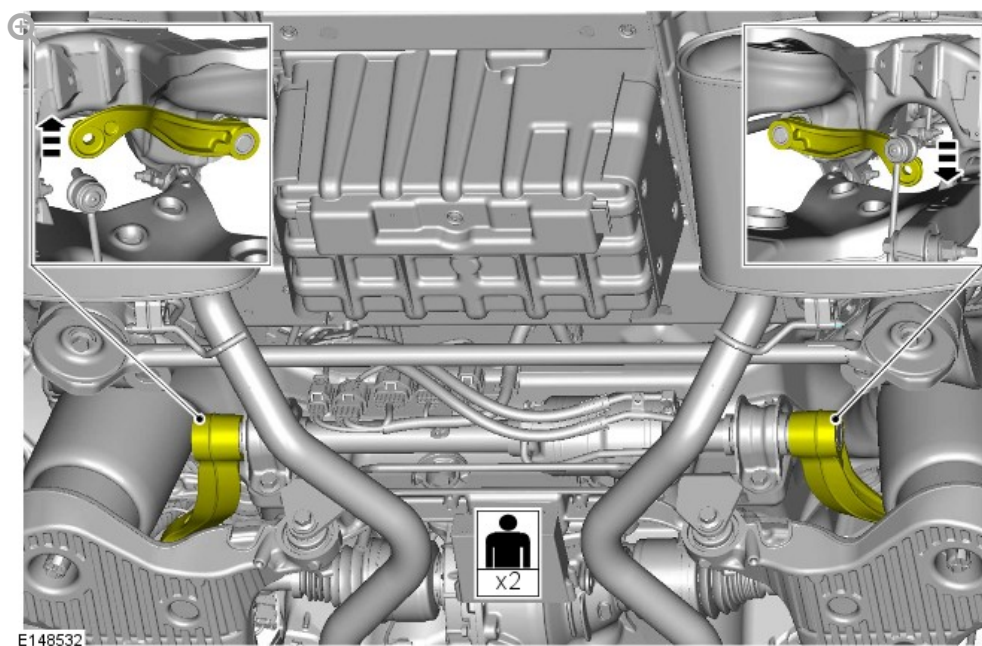
NOTES:

- The drop link fixing removal must be carried out on both sides.
- RH illustration shown, LH is similar.



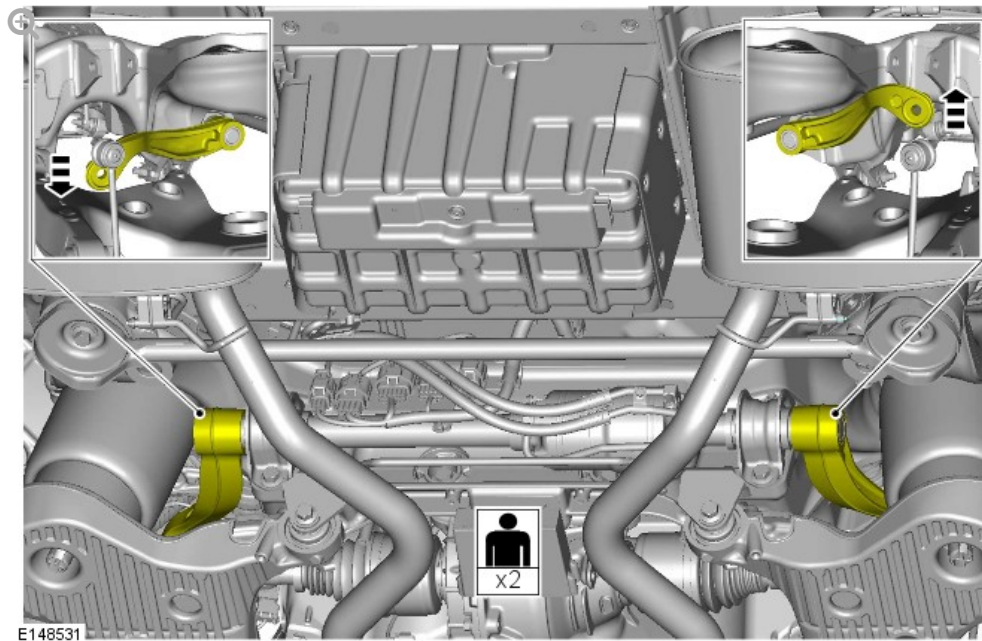
Bleed rear actuator:

22.



Open right bleed screw and right arm fully down and left arm up.
Allow fluid to run for a further 5 seconds. Close bleed screw.

23.

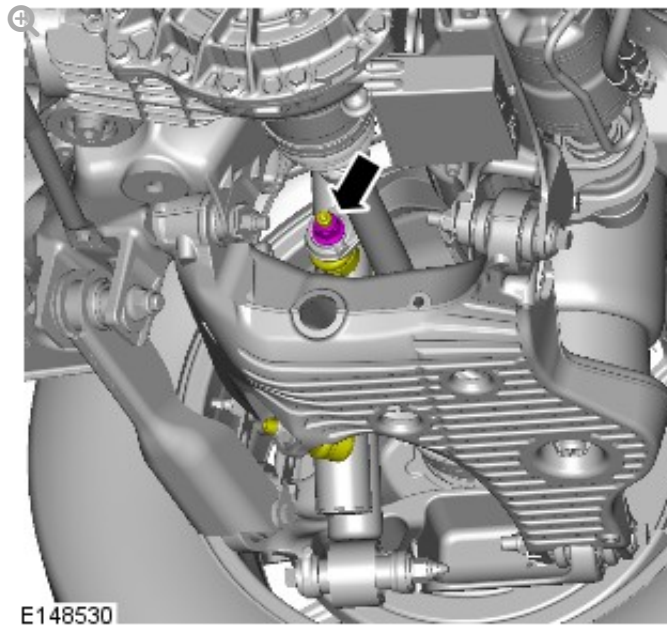


Open left bleed screw and right arm fully up and left arm down.
Allow fluid to run for a further 5 seconds. Close bleed screw.

24.

NOTES:

- RH illustration shown, LH is similar.
- The step must be carried out on both sides.



Torque: 185 Nm

25.

WARNING:

The special tool is still pressurised when the source air pressure is removed. Release air pressure within special tool slowly before removing. Remove the special tool from the dynamic response system reservoir.

Stop the manual bleed in approved diagnostic tool and check the bleed.

26.

1. Check front bleed:
2. Disconnect only one front stabilizer link. Stroke the arm up and down free play travel should be less than $\pm 10\text{mm}$. If greater than $\pm 10\text{mm}$ travel observed and/or spongy feeling a system re-bleed will be required.

27.

1. Check rear bleed:
2. Disconnect only one rear stabilizer link. Stroke the arm up and down free play travel should be less than $\pm 10\text{mm}$. If

greater than $\pm 10\text{mm}$ travel observed and/or spongy feeling
a system re-bleed will be required.

28. Repeat steps 14-23 if required.
29. Torque front and rear stabilizer links to the required torque. Tighten the bleed screw to 15 Nm.
30. Install the bleed screw covers.
31. Refer to: [Engine Undershield](#) (501-02 Front End Body Panels, Removal and Installation).
32. Make sure that the fluid level is set to maximum line.
33. If the valve block has been replaced run the auto bleed procedure from within SDD to release air that may be trapped in the block. This requires the engine to be running. Using SDD clear pressure sensor adaptive data.
34. Using approved diagnostic tool carry out hydraulic system response test to confirm correct system operation.
35. Drive the vehicle for approximately 15 minutes performing manoeuvres which energise the system and include wheel inputs.